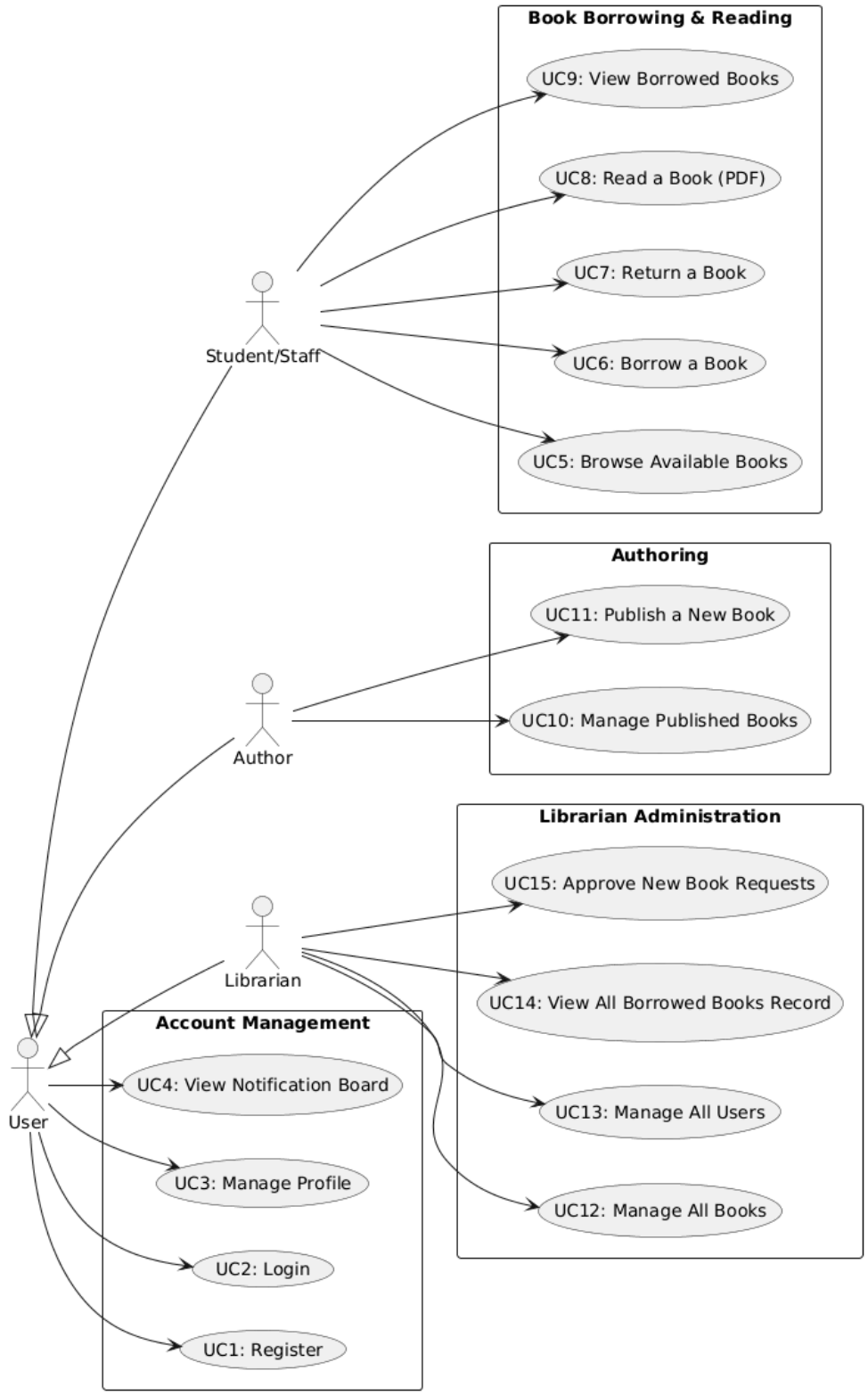
HUI, Ka Kuen 20889062 (Responsible for Book Borrowing & Reading)

TSANG, Tsz Hin 21055553 (Responsible for Account Management, Authoring)

YAU, Man Kit Bosco 21039341 (Responsible for Librarian Administration)

Use Case Diagram:



Package for User: Account Management

Use Case 1 – Register

Brief Description
This use case describes how a new user (Student/Staff, Author, or Librarian) creates an account in the system.

Basic Flow

1. The use case begins when the user chooses to register.
2. The system displays the registration form, requesting:
    - Username
    - Full Name
    - Email
    - Password
    - Role (Student/Staff, Author, Librarian)
3. The user enters the required information and submits.
4. The system validates all inputs:
    - Username uniqueness across all user types
    - Full name not empty
    - Password strength (strong, weak, or empty)
5. The system creates a new user account with the specified role and stores it in the database.
6. The system notifies the user that registration was successful and prompts them to log in.
7. The use case ends.

Alternative Flows

A1: Username Already Exists
At step 4, if the username is already taken by any existing user (regardless of role):

1. The system displays an error message: “Username is already taken. Please choose a different username.”
2. The flow of events is resumed at step 2 of the Basic Flow.

A2: Full Name Empty
At step 4, if the Full Name field is left blank:

1. The system displays an error message: “Full Name cannot be empty.”
2. The flow of events is resumed at step 2 of the Basic Flow.

A3: Password Validation
At step 4, the system checks the password strength:

- A3.1: Password Empty – If the password field is blank:

    1. The system displays an error message: “Password cannot be empty.”
    2. The flow resumes at step 2.
- A3.2: Weak Password – If the password does not meet complexity requirements (e.g., at least 8 characters, includes uppercase, lowercase, digit, special character):
    1. The system displays an error message: “Password is weak. Please use at least 8 characters including uppercase, lowercase, a number, and a special character.”
    2. The flow resumes at step 2.

A4: Email Already Exists (additional, if needed)
At step 4, if the email is already associated with an existing account:

1. The system displays an error message: “Email is already registered.”
2. The flow of events is resumed at step 2.

Use Case 2 – Login

Brief Description
This use case describes how a registered user logs into the system.

Basic Flow

1. The use case begins when the user chooses to log in.
2. The system displays the login interface, requesting:
    - Username
    - Password
    - Role (Student/Staff, Author, Librarian)
3. The user enters their credentials and role, then submits.
4. The system validates:
    - The username exists and matches the selected role.
    - The password is correct for that user.
5. The system grants access and navigates the user to their personalized dashboard.
6. The use case ends.

Alternative Flows

A1: Invalid Credentials
At step 4, if the username does not exist or the password is incorrect:

1. The system displays an error message: “Invalid credentials.”
2. The flow of events is resumed at step 2 of the Basic Flow.

A2: Role Mismatch
At step 4, if the username exists but its associated role does not match the role selected by the user:

1. The system displays an error message: “Invalid credentials.”
2. The flow of events is resumed at step 2 of the Basic Flow.

Use Case 3 – Manage Profile

Brief Description
This use case allows any logged-in user to view and update their profile information. Depending on the user's role, additional fields may be editable (bio for authors, employee ID for librarians). Password changes require re-authentication and real-time strength feedback.

Basic Flow

1. The use case begins when the user selects "Manage Profile".
2. The system retrieves and displays the user's current profile information, including role-specific editable fields:
    - All users: Full Name, Password, etc.
    - Author: additionally a "Bio" field.
    - Librarian: additionally an "Employee ID" field.
3. The user modifies one or more fields.
4. If the password field is changed, the system performs subflow S1.1: Password Strength Meter in real time as the user types.
5. The user submits the changes.
6. The system performs subflow S1.2: Re-authenticate Password (requires current password).
7. If re-authentication succeeds, the system validates all updated credentials (e.g., Full Name not empty, password meets strength requirements, email format).
8. The system updates the database with the new information.
9. The system displays a confirmation message: "Profile updated successfully."
10. The use case ends.

Subflows

S1.1: Password Strength Meter (real-time)

1. As the user types a new password, the system evaluates its strength (e.g., weak, medium, strong) based on length, character variety, etc.
2. The system displays a visual indicator (e.g., a colored bar or text) showing the current strength.
3. If the password is empty or weak, the system may show a hint message (e.g., "Use at least 8 characters, including uppercase, lowercase, number, and special character").
4. This subflow repeats dynamically until the user finishes typing.

S1.2: Re-authenticate Password

1. The system displays a dialog prompting the user to enter their current password.
2. The user enters the current password.
3. The system verifies the password against the stored credential.
4. If correct, the subflow ends successfully.

5. If incorrect, the system displays an error message and re-prompts (up to 3 attempts before aborting).

Alternative Flows

A1: Re-authentication Fails
At step 6 (subflow S1.2), if the user fails to provide the correct current password after the allowed attempts:

1. The system displays an error message: “Re-authentication failed. No changes were saved.”
2. The use case ends without updating the profile.

A2: Validation Fails for Updated Fields
At step 7, if any validation rule is violated (e.g., Full Name empty, new password too weak, email invalid):

1. The system displays specific error messages for each invalid field.
2. The user is returned to the edit screen (step 2) with previously entered values preserved.
3. The use case does not proceed to update.

A3: No Changes Detected
At step 5, if the user submits the form without modifying any field:

1. The system informs the user: “No changes were made.”
2. The use case ends.

A4: Author Updates Bio – No Additional Validation
Bio field may have a length limit (e.g., 500 characters) – if exceeded, the system shows an error and returns to edit.

A5: Librarian Updates Employee ID – Uniqueness Check
If the system requires employee IDs to be unique across librarians, the system validates that the new ID is not already in use. If duplicate, an error message is shown and the flow returns to edit.

Use Case 4 – View Notification Board

Brief Description
This use case allows any logged-in user to view system notifications relevant to their role. Notifications are timestamped, categorized, and support actions such as marking as read, archiving, searching, filtering, and deleting.

Basic Flow

1. The use case begins when the user selects “Notification Board”.
2. The system retrieves the user’s role (Student/Staff, Author, or Librarian).
3. The system displays the notification board interface, which includes:
    - A list of notifications specific to the user’s role (see Role-Specific Notifications below).
    - An unread notification counter (e.g., “You have 3 unread notifications”).
    - Options to filter by category (e.g., “Due Reminders”, “Approval Updates”, “Announcements”).
    - A search bar (by keyword).
    - For each notification: timestamp, category, content, and actions (mark as read, archive, delete).
4. The user may perform any of the following activities:
    4.1. View notifications – scrolling through the list.
    4.2. Mark as read – select a notification and mark it read; the unread counter decreases.
    4.3. Archive notifications – move selected notifications to an archive folder (hidden from main view).
    4.4. Delete notifications – permanently remove selected notifications.
    4.5. Search/filter – refine the list by category or keyword.
5. The system updates the notification statuses (read/unread, archived, deleted) in the database.
6. The use case ends when the user exits the notification board.

Role-Specific Notifications

User Role	Notification Types	Examples
Student/Staff	– Book due reminders – Book deletion notice (only for users who borrowed that specific book) – Other announcements	“Your borrowed book ‘Software Engineering’ is due in 2 days.” “The book ‘Data Structures’ you borrowed has been removed from the library.”
Author	– Book approval/rejection updates – Other announcements	“Your book ‘AI Basics’ has been approved and is now available.” “Your book ‘Poetry Collection’ was rejected. Reason: Copyright issue.”
Librarian	– New book submissions (pending approval) – User account updates (e.g., new registrations, profile changes) – Other announcements	“A new book ‘Machine Learning’ has been submitted by author [author name].” “User ‘alice’ updated their profile.”

All notifications include a timestamp (date and time of creation) and a category (e.g., “Due Reminder”, “Deletion Notice”, “Approval Update”, “New Submission”, “Announcement”).

Alternative Flows

A1: No Notifications
If the user has no notifications (including archived ones), the system displays a message: “No notifications to display.”

A2: Search Returns No Results
If a search or filter yields an empty list, the system displays: “No notifications match your criteria.” The user may clear filters or search terms.

A3: Attempt to Delete Read-Only Notification
(If any notification is marked as system-critical and non-deletable – optional) The system informs the user that certain notifications cannot be deleted; they may only be archived.

A4: Archive Instead of Delete
If the user attempts to delete a notification that should be retained for legal/audit reasons, the system may automatically archive it instead and inform the user.

Package for Student/Staff: Book Borrowing & Reading

Use Case 5 – Browse Available Books

Brief Description
This use case allows a Student/Staff to search, filter, and browse books that are currently available in the library, and to view a book's summary before deciding to borrow.

Basic Flow

1. The use case begins when the user selects "Browse Available Books".
2. The system displays a search and filter interface with options:
    - Search by title, author, or ISBN
    - Filter by genre, publish date range, or availability (available/borrowed)
3. The user enters search criteria and/or selects filters, then submits.
4. The system retrieves and displays a list of books matching the criteria.
5. The user may select a book from the list to view its details.
6. The system displays the book's full details, including:
    - Title, author, genre, publish date, availability status
    - Summary (description/abstract)
    - If the summary exceeds a length threshold, the system provides a "Read More" option that opens a pop-up screen for a quick, scrollable view.
7. The user may close the details view and optionally refine the search/filters (return to step 2).
8. The use case ends when the user exits the browse interface.

Alternative Flows

A1: No Results Found
At step 4, if no books match the search/filter criteria:

1. No books will be displayed.
2. The user may refine the search/filters (return to step 2).

A2: Book Summary Unavailable
At step 6, if the book has no summary (e.g., for older records without generated summaries):

1. The system displays a placeholder: "No summary available."
2. The user may still proceed with borrowing (via a separate action) or close the details.

Use Case 6 – Borrow a Book

Brief Description
This use case describes how a Student/Staff borrows one or more books from the library, subject to borrowing limits and duration constraints.

Basic Flow

1. The use case begins when the user chooses to borrow books.
2. Perform subflow S2: Select Books (updated to allow multiple selections).
3. For each selected book, the system checks if the book is available.
4. after this transaction and verifies it does not exceed the maximum borrowing limit (e.g., 5 books).
5. The system prompts the user to select a borrowing duration (in days) within the allowed range (e.g., 1 to 14 days).
6. The user specifies the duration.
7. The system displays a confirmation summary listing all selected books, the due date (calculated from today + selected duration), and the total count.
8. The user confirms the borrowing.
9. The system records each borrowing transaction in the database, updates each book's status to "borrowed", and notifies the user of successful borrowing.
10. The use case ends.

Subflows

S2: Select Books (Updated)

1. The user may search/browse available books (Use Case 5).
2. The user selects one or more books to borrow (e.g., by checking checkboxes).
3. The user proceeds to confirm the selection.

Alternative Flows

A1: Borrowing Limit Exceeded
At step 4, if the total number of books the user would have after this transaction exceeds the maximum limit (e.g., currently has 3 books and tries to borrow 3 more, exceeding a limit of 5):

1. The system displays an error message: "You cannot borrow more than [X] books at a time. You currently have [Y] books borrowed. You may borrow up to [X-Y] additional books."
2. The user is returned to the selection screen (subflow S2) to adjust the selection.

A2: Invalid Borrowing Duration
At step 5, if the user enters a duration outside the allowed range (e.g., less than minimum or greater than maximum):

1. The system displays an error message: “Borrowing duration must be between [min] and [max] days.”
2. The flow returns to step 5 to allow re-entry.

Use Case 7 – Return a Book

Brief Description
This use case allows a Student/Staff to return borrowed books. Returns can be performed manually by the user (self-return) or automatically by the system when the borrowing period expires.

Basic Flow (Self-return)

1. The use case begins when the user selects “Return a Book”.
2. The system retrieves and displays the list of books currently borrowed by the user.
3. The user selects one or more books to return.
4. The user confirms the return.
5. The system updates the borrowing records, marks each book as “available”, and calculates any overdue fines (if applicable).
6. The system notifies the user of successful return.
7. The use case ends.

Auto-return Flow (Triggered by System)

1. The use case begins automatically when the borrowing period for a book expires (detected by a scheduled job or real-time check).
2. The system identifies all overdue books that have not been manually returned.
3. For each such book:
   3.1. The system updates the borrowing record to mark the book as returned.
   3.2. The system marks the book as “available” in the database.
   3.3. The system calculates any overdue fines (if applicable) and adds them to the user’s account.
4. The system sends a notification to the user: “Your borrowing period for [book title] has expired. The book has been automatically returned. Any overdue fines have been applied.”
5. The use case ends.

Alternative Flows (Self-return)

A1: No Borrowed Books
If the user has no borrowed books when trying to self-return, the system displays a message and the use case ends.

A2: Return Confirmation Cancelled
If the user cancels the confirmation step, the use case ends without any changes.

A3: Overdue Fine Calculation
If a book is returned (either self-return after due date or auto-return), the system calculates fines based on the number of days overdue. The user is shown the fine amount in the notification.

Use Case 8 – Read a Book

Brief Description
This use case describes how a Student/Staff reads a borrowed book's PDF content. The system adds a dynamic watermark, and the user can save bookmarks (reading progress) and highlight text for future reference.

Basic Flow

1. The use case begins when the user selects "Read" for a borrowed book.
2. The system retrieves the PDF file.
3. The system dynamically adds a watermark (user name, date) to each page.
4. The system displays the PDF in an interactive reader interface with the following features:
    - Navigation (next/previous page, go to page)
    - Bookmark management (add, view, delete bookmarks)
    - Text highlighting (select text and apply highlight)
5. While reading, the user may:
   5.1. Perform subflow S3: Manage Bookmarks (save reading progress).
   5.2. Perform subflow S9: Manage Highlights (add, view, remove highlights).
6. The user exits the reader.
7. The system saves any unsaved bookmarks or highlights automatically (or upon exit).
8. The use case ends.

Subflows

S3: Manage Bookmarks

1. The user may add a bookmark at the current page.
2. The system prompts for a bookmark name (optional).
3. The user enters a name (or uses default "Page X").
4. The system saves the bookmark with the page number and timestamp.
5. The user may also view a list of existing bookmarks for this book and click to jump to that page.
6. The user may delete existing bookmarks.

S9: Manage Highlights

1. The user selects a portion of text in the PDF (by clicking and dragging).
2. A toolbar appears with a "Highlight" option.
3. The user clicks "Highlight"; the system applies a background color (e.g., yellow) to the selected text.
4. The system saves the highlight coordinates and the selected text snippet in the user's profile for this book.
5. The user may view all highlights for this book (e.g., in a sidebar), click on a highlight to jump to that location, or remove a highlight.

Alternative Flows

A1: PDF Rendering Error
If the PDF file is corrupted or cannot be displayed, the system shows an error message: “Unable to load the book. Please contact the librarian.” The use case ends.

A2: Highlighting Not Supported on This PDF
If the PDF is scanned/image-based and text selection is unavailable, the system disables the highlight feature and notifies the user: “Text highlighting is not available for this book.”

A3: Borrowing Period Expired
At step 2 or during the periodic check, if the borrowing period has ended:

1. The system displays a warning message: “Your borrowing period for this book has expired. The reading screen will now close.”
2. The system automatically saves any unsaved bookmarks/highlights.
3. The system closes the reader interface.
4. The system triggers the auto-return process (as in UC7 – Return a Book).
5. The use case ends.

UC9 – View Borrowed Books

Brief Description
This use case allows a Student/Staff to view their currently borrowed books and to search/filter that list by title, author, genre, publish date, or availability status (i.e., borrowed/overdue).

Basic Flow

1. The use case begins when the user selects “View Borrowed Books”.
2. The system retrieves the list of books the user has currently borrowed (not returned).
3. The system displays a search and filter interface with options:
    - Search by title or author
    - Filter by genre, publish date range, or availability (e.g., “borrowed” or “overdue”)
4. The user enters search criteria and/or selects filters, then submits.
5. The system filters the borrowed book list accordingly and displays the results.
6. For each borrowed book, the system shows:
    - Title, author, genre, publish date
    - Due date
    - Status (e.g., “on time” or “overdue”)
7. The user may select a book to view more details or to perform actions (e.g., return, read, renew – covered in other use cases).
8. The user may refine the search/filters (return to step 3) or exit.
9. The use case ends.

Alternative Flows

A1: No Borrowed Books
If the user has no currently borrowed books, the system displays a message: “You have not borrowed any books at this time.”

A2: Search/Filter Returns No Results
If the search/filter criteria yield no matches among the user’s borrowed books, the system displays: “No borrowed books match your criteria.” The user may clear filters and try again.

A3: Book Overdue
If a borrowed book’s due date has passed, its status is shown as “overdue” and may be highlighted (e.g., in red). The system may also provide a link to return or renew the book.

Package for Author: Authoring

Use Case 10 – Manage Published Books

Brief Description
This use case allows an author to view, read, modify, or delete books they have published or submitted. The system shows each book's status (Approved/Rejected/Pending). Modification is allowed only if the book is pending approval, or if it is published and not currently borrowed by any student/staff.

Basic Flow

1. The use case begins when the author selects "Manage Published Books".
2. The system retrieves and displays a list of all books associated with the author, showing for each:
    - Title
    - Genre
    - Status (Approved, Rejected, or Pending)
3. The author may select a book from the list.
4. The system presents available actions based on the book's status and borrowing state:
   4.1. View Details – always available (displays metadata, summary, statistics).
   4.2. Read Book – always available (opens the PDF reader with watermark; no borrowing check because it's the author's own book).
   4.3. Modify – available only if the book is pending approval, OR (published and not currently borrowed by any user). Otherwise, the option is disabled or shows an error.
   4.4. Delete – available only if the book is not currently borrowed (any status).
5. Depending on the selected action:
   5.1. If View Details – perform subflow S4: View Book Details.
   5.2. If Read Book – perform subflow S11: Read Own Book.
   5.3. If Modify – perform subflow S5: Modify Book Details with restrictions.
   5.4. If Delete – perform subflow S6: Delete Book with confirmation.
6. After completing the action, the system returns to the list (step 2).
7. The use case ends.

Subflows

S4: View Book Details (unchanged)
Displays full metadata, LLM summary, statistics.

S11: Read Own Book

1. The system retrieves the PDF file (for both published and unpublished books).
2. The system dynamically adds a watermark (author's name, "Author Copy" or similar).
3. The system displays the PDF in a reader interface (without borrowing period checks, as it's the author's own work).
4. The author may use reading features (bookmarks, highlights – same as UC8).

5. The author exits the reader.

S5: Modify Book Details (Restricted)

1. The system checks if modification is allowed:
    - If book status is Pending → allowed.
    - If book status is Approved (published) and not borrowed → allowed.
    - If book status is Rejected → not allowed.
    - If book status is Approved but currently borrowed → not allowed.
2. If not allowed, the system displays an error message (see A1 or A2) and returns to the list.
3. If allowed, the system displays editable fields (title, genre, description, etc.).
4. The author makes changes and confirms.
5. The system validates the changes (e.g., title not empty).
6. The system updates the database and notifies the author of success.

S6: Delete Book (with Confirmation)

1. The system checks if the book is currently borrowed (any status). If borrowed, it triggers Alternative Flow A3.
2. Otherwise, the system displays a confirmation dialog: “Are you sure you want to delete ‘[book title]’? This action cannot be undone.”
3. The author confirms.
4. The system removes the book entry from the database.
5. The system notifies the author of successful deletion.

Alternative Flows

A1: Modify Not Allowed – Rejected Book
At subflow S5 step 1, if the book status is “Rejected”:

1. The system displays an error message: “Cannot modify a rejected book. Please resubmit as a new book.”
2. The flow returns to the book list.

A2: Modify Not Allowed – Borrowed Book
At subflow S5 step 1, if the book is approved (published) and currently borrowed by any student/staff:

1. The system displays an error message: “Cannot modify this book because it is currently borrowed. Please wait until all copies are returned.”
2. The flow returns to the book list.

A3: Delete Not Allowed – Borrowed Book
At subflow S6 step 1, if the book is currently borrowed (any status):

1. The system displays an error message: “Cannot delete a book that is currently borrowed. Please wait until it is returned.”
2. The flow returns to the book list.

A4: Validation Fails During Modify
If the author submits invalid data (e.g., empty title), the system shows an error and returns to the edit screen.

Use Case 11 – Publish a New Book

Brief Description
This use case allows an author to upload a new book, after which an LLM generates a summary.

Basic Flow

1. The use case begins when the author selects “Publish New Book”.
2. The system displays a submission form (metadata + PDF upload).
3. The author enters metadata and uploads the PDF.
4. The author submits.
5. The system validates the file format.
6. The system creates a book entry with status “pending approval”.
7. The system invokes an LLM to generate a summary and stores it.
8. The system notifies the author that the book is pending librarian approval.
9. The use case ends.

Alternative Flows

A1: Invalid File Format
At step 5, if the uploaded file is not a PDF, the system informs the author and resumes at step 2.

Package for Librarian: Librarian Administration

Use Case 12 – Manage All Books

Brief Description
This use case allows a librarian to add, modify, or delete any book in the system, regardless of author.

Basic Flow

1. The use case begins when the librarian selects “Manage All Books”.
2. The system displays a searchable list of all books.
3. While the librarian has an activity to perform:
   3.1. If Add Book is selected:
   3.1.1. Perform subflow S7: Add Book.
   3.2. If Modify is selected for a book:
   3.2.1. Perform subflow S5: Modify Book Details (reused).
   3.3. If Delete is selected for a book:
   3.3.1. Perform subflow S6: Delete Book (reused).
4. The use case ends.

Subflows

S7: Add Book

1. The system displays a form for entering book metadata (title, author, ISBN, genre, etc.).
2. The librarian enters the details and submits.
3. The system validates ISBN uniqueness.
4. The system creates a new book entry with status “available”.
5. The system notifies the librarian of success.

Use Case 13 – Manage All Users

Brief Description
This use case allows a librarian to view, edit, or deactivate user accounts (students, staff, authors, librarians). The librarian can perform actions on individual accounts or in bulk, and view activity logs for each user (e.g., last login, number of borrowed books).

Basic Flow

1. The use case begins when the librarian selects “Manage All Users”.
2. The system displays a list of all registered users with search/filter options (by name, role, status).
3. For each user, the system shows:
    - Username, full name, role, account status (active/deactivated)
    - Activity log (last login timestamp, number of currently borrowed books)
4. The librarian may select one or more users (checkboxes).
5. While the librarian has an activity to perform:
   5.1. If View/Edit is selected for a single user:
   5.1.1. Perform subflow S12: Edit Single User.
   5.2. If Deactivate is selected for one or more users:
   5.2.1. Perform subflow S13: Bulk Deactivate Users.
   5.3. If Bulk Update is selected (e.g., change role or status for multiple users):
   5.3.1. Perform subflow S14: Bulk Update Users.
6. The system confirms all changes and updates the database.
7. The use case ends.

Subflows

S12: Edit Single User

1. The system displays the user’s full profile (editable fields: full name, role, account status, etc.).
2. The librarian modifies the desired fields.
3. The system prompts for confirmation: “Are you sure you want to save these changes?”
4. The librarian confirms.
5. The system validates the changes (e.g., role exists, status valid).
6. The system updates the database and displays a success message.

S13: Bulk Deactivate Users

1. The system lists the selected users and asks for confirmation: “Deactivate [N] selected user accounts? Deactivated users cannot log in.”
2. The librarian confirms.
3. The system updates each selected user’s status to “deactivated”.
4. The system displays a success message and notifies affected users (optional).

S14: Bulk Update Users

1. The system prompts the librarian to choose an action to apply to all selected users (e.g., change role to “Author”, set status to “active”).
2. The librarian selects the action and provides any required parameters.
3. The system displays a confirmation dialog summarizing the changes.
4. The librarian confirms.
5. The system validates the action for each user (e.g., cannot change a librarian’s role to student if that would break permissions).
6. The system updates the database for all selected users.
7. The system displays a success message with the number of affected users.

Alternative Flows

A1: No Users Selected for Bulk Action
If the librarian attempts a bulk action without selecting any users, the system displays an error message: “Please select at least one user.” The flow returns to the list.

A2: Validation Fails for Bulk Update
If the bulk update is invalid for some users (e.g., trying to deactivate the only head librarian), the system shows an error listing the problematic users and does not apply any changes. The librarian may adjust the selection.

A4: User Not Found
If a search yields no users, the system displays a message and allows the librarian to

Use Case 14 – View All Borrowed Books Record

Brief Description
This use case allows a librarian to view a complete record of all books currently borrowed across the entire system. The record includes key details and supports search and filtering for efficient management.

Basic Flow

1. The use case begins when the librarian selects “View All Borrowed Books Record”.
2. The system retrieves all active borrowing records (books that have been borrowed but not yet returned).
3. The system displays a table with the following columns for each record:
    - Book Title
    - Borrower Username
    - Borrow Date
    - Return Date (due date, or actual return date if returned early)
    - Status (e.g., “Borrowed – On Time”, “Borrowed – Overdue”, “Returned”)
4. The system provides a search and filter interface with options:
    - Search by Book Title (partial match)
    - Search by Borrower Username (partial match)
    - Filter by Status (e.g., “Overdue”, “On Time”, “Returned”)
    - Filter by Date Range (Borrow Date or Return Date)
5. The librarian may enter search criteria and/or select filters, then submit.
6. The system filters the records accordingly and refreshes the display.
7. The librarian may clear filters to see all records again.
8. The use case ends when the librarian exits the screen.

Alternative Flows

A1: No Borrowed Records
If there are no active borrowing records in the system, the system displays a message: “No borrowed books at this time.”

A2: Search/Filter Returns No Results
If the search/filter criteria yield no matching records, the system displays: “No records match your criteria.” The librarian may clear filters and try again.

A3: Return Date Not Yet Set
For active borrowings (not yet returned), the Return Date column shows the due date. Status indicates “Borrowed – On Time” or “Borrowed – Overdue” based on current date.

A4: Returned Books
For books that have already been returned (if the librarian wishes to include historical records as an option), the system may provide a toggle to include or exclude returned books. The

default is to show only currently borrowed books. Returned records show the actual return date and Status “Returned”.

Use Case 15 – Approve New Book Requests

Brief Description
This use case allows a librarian to review and approve or reject books submitted by authors. When rejecting, the librarian must provide a brief reason, which is stored and sent as feedback to the author.

Basic Flow

1. The use case begins when the librarian selects “Approve New Book Requests”.
2. The system displays a list of books with status “pending approval”.
3. The librarian selects a pending book.
4. The system displays the book’s metadata and the PDF for review.
5. While the librarian has an activity to perform:
   5.1. If Approve is selected:
   5.1.1. The system updates the book status to “available”.
   5.1.2. The system notifies the author of approval.
   5.2. If Reject is selected:
   5.2.1. Perform subflow S8: Provide Rejection Reason.
   5.2.2. The system updates the book status to “rejected” and stores the reason.
   5.2.3. The system notifies the author of rejection, including the rejection reason.
6. The use case ends.

Subflows

S8: Provide Rejection Reason

1. The system displays a text area for entering a rejection reason.
2. The librarian enters a brief explanation (required).
3. The librarian confirms the rejection.

Alternative Flows

A1: Rejection Reason Empty
At step 5.2.1 (subflow S8), if the librarian attempts to submit without entering a reason:

1. The system displays an error message: “A rejection reason is required.”
2. The librarian is prompted to enter a reason; the flow returns to step 1 of subflow S8.